

MOHAMMAD NURULLAH PATWARY

✉ noor.fahim1192@gmail.com  [linkedin.com/in/nurullah-patwary](https://www.linkedin.com/in/nurullah-patwary)  github.com/darkprince92
Arlington, Texas, US ◇ PhD Candidate, UT Arlington ◇ Advisor: Dr. Allison Sullivan

RESEARCH INTERESTS

Natural Language Processing · Language Models & Prompting · NL-to-Formal Specification · Program Synthesis from Natural Language · Human-AI Interaction · Interactive Developer Tools · Mutation Testing

EDUCATION

University of Texas at Arlington, Arlington, Texas *Aug 2023 – Present*

PhD in Computer Science CGPA: 4.0 Expected: May 2028

Advisor: Dr. Allison Sullivan Research: Formal methods usability, NLP-assisted specification, live programming

Bangladesh University of Engineering and Technology, Dhaka, Bangladesh *Jan 2010 – Sep 2015*

BS in Computer Science and Engineering

RESEARCH EXPERIENCE

University of Texas at Arlington, Arlington, Texas

Graduate Research Assistant *May 2024 – Present*

- Refined Alloy mutation operators via empirical analysis of common user errors, making mutation testing more representative of real-world faults; published at **FormalISE 2025** (IEEE/ACM). [\[DOI\]](#)
- Developed **formula completion for Alloy** using syntactic and semantic analysis to synthesize valid continuations for partial logical constraints—analogue to code completion for formal specification languages; under review 2025.
- Built a **VS Code language server extension (LSP)** for Alloy in Java (Lsp4j), providing context-aware code completion, syntax highlighting, and error reporting for formal-methods practitioners.
- Investigating **LLM-assisted specification generation**: prompting large language models to draft and repair Alloy constraints from informal natural-language requirements, bridging NLP and formal logic.

PUBLICATIONS

- Jovanovic, A.; **Patwary, M. N.**; Sullivan, A. *Refining Alloy-Based Mutation Operators to Reflect Common Mistakes*. **FormalISE 2025** (IEEE/ACM). [\[DOI\]](#)
- Patwary, M. N.**; Jovanovic, A.; Sullivan, A. *A Structure Editor for Building Software Models*, arXiv 2024. [\[arXiv\]](#)
- Patwary, M. N.**; Uddin, I.; Sullivan, A. *Formula Completion Suggestions for Alloy Models*. Under review, 2025.

SKILLS

NLP / ML	Transformers (HuggingFace), prompt engineering, fine-tuning, text classification, seq2seq
Languages	Python, Java, JavaScript
ML Frameworks	PyTorch, scikit-learn, spaCy, NLTK
Dev Tools	VS Code LSP, Lsp4j, Spring, Django
Infrastructure	AWS (S3, Lambda, DynamoDB, EC2), Docker, Apache Kafka, Redis, PostgreSQL, Elasticsearch

INDUSTRY EXPERIENCE

Agoda Services Co., Ltd., Bangkok, Thailand

Software Engineer *Jun 2022 – Jul 2023*

- Engineered backend authentication service for a global travel platform (tens of millions of daily users); analyzed login-event streams to detect credential-stuffing patterns, shipped 2FA validated via A/B testing.
- Strengthened ELK-based monitoring pipeline for real-time login-attack detection across segregated security zones.

Priyo Sys Ltd., Dhaka, Bangladesh

Senior Software Engineer *Feb 2020 – May 2022*

- Architected a nationwide quiz platform on AWS (S3, Lambda, DynamoDB, Redis) supporting 100K concurrent users and 50M pre-generated questions for a government-commissioned national event; ran without incident at scale.
- Extended Django ORM with field-level access control and row-level versioning, enforcing hierarchical permissions transparently at the data layer for an enterprise people-data inventory product.

iPay Systems Ltd., Dhaka, Bangladesh

Software Engineer *Dec 2015 – Dec 2019*

- Built event-driven promotional rule engine consuming Kafka events and evaluating composable Groovy DSL rules; daily sign-ups grew from 150 to 30,000 (200×), driven by flexible promotional programs this engine enabled.
- Automated user verification via OCR and face recognition against national ID database, reducing onboarding from hours to minutes; built Kafka Connect streaming data warehouse (Elasticsearch / PostgreSQL) for analytics.

HONORS & ACHIEVEMENTS

- Top 10 distinguished poster award, undergraduate thesis presentation, BUET CSE, 2015.
- Top project, Water Hackathon App Fest Bangladesh 2014 (World Bank).
- 1st place, Database Project Showcase, BUET CS&E, 2013.